Math 340 • Fall 2025 • Victor Matveev

Polynomial Interpolation Error

One big advantage of Newton's divided difference method of writing down interpolating polynomial is the ease with which we can analyze its error. In fact, the remainder (error) of interpolation looks exactly like an extra interpolation term obtained by adding one more point:

$$f(x) = \underbrace{f[x_1] + f[x_1,x_2](x-x_1) + \ldots + f[x_1,\ldots,x_n](x-x_1)\ldots(x-x_{n-1})}_{P_{n-1}(x)} + \underbrace{f[x_1,\ldots,x_n,x](x-x_1)\ldots(x-x_n)}_{\text{Remainder / Error, } R_{n-1}(x)}$$

The **proof** of this remainder formula is remarkably simple: consider x in the above expression as another interpolation node, $x \equiv x_{n+1}$; then the above is simply a formula for $P_n(x)$ -- an interpolant with one extra point -- which obviously exists for any value of $x \equiv x_{n+1}$.

If can also be elegantly proven that each of the coefficients above is related for the derivative of $f(x)$, highlighting the analogy between interpolating polynomial and a "distributed" form of Taylor expansion:

$$\exists c \in \text{Interval containing all } \{x_k\}_{k=1}^n, \text{ such that } f[x_1,\ldots,x_n] = \frac{f^{(n-1)}(c)}{(n-1)!}$$

For the remainder, $\boldsymbol{R_{n-1}(x) = f[x_1,\ldots,x_n,x](x-x_1)\ldots(x-x_n)}$, this implies that

$$\exists c \in \text{Interval containing all } \{x_k\}_{k=1}^n \text{ and } x, \text{ such that } R_{n-1}(x) = \frac{f^{(n)}(c)}{n!}(x-x_1)\ldots(x-x_n)$$

Even though we don't know where the argument value ***c*** resides, the latter expression is useful when finding the upper bound on the error. Let's consider a simple example to examine these two ways of analyzing interpolation error:

EXAMPLE: Consider **linear** approximation of $f(x)$ using nodes $\{x_k\} = \{-1,1\}$, and examine its error over interval [-1, 1], in two different ways:

1) Compute the error in the middle of the interval, at $x = 0$ (since we expect the error to be largest near that point):

2) Analyze the error bound using the formula for the coefficients given above.

Frist, let's find the interpolating polynomial (which is a line in this case of just two interpolation nodes).

Since we also want to find $\boldsymbol{R_1(0) = f[x_1,x_2,0](x-x_1)(x-x_2)}$, let's add a row for the extra value $\boldsymbol{x = 0}$:

x_k	$f[x_k]$	$f[x_{k-1},x_k]$	$f[x_1,x_2,0]$
-1	e^{-1}		
1	e	$\frac{e-e^{-1}}{1-(-1)} = \frac{e-e^{-1}}{2} = \sinh(1)$	
0	1	$\frac{1-e}{0-1} = e-1$	$\frac{e-1-\frac{e-e^{-1}}{2}}{0-(-1)} = \frac{e+e^{-1}}{2} - 1 = \cosh 1 - 1$

Note : it is no surprise that these coefficients are exactly the ones we found in the homework (with points in a different order though)!

Thus, our interpolating line is:

$$f(x) \sim P_1(x) = f[x_1] + f[x_1,x_2](x-x_1) = e^{-1} + \sinh(1)(x+1) = \underbrace{e^{-1} + \sinh(1)}_{\cosh(1)} + \sinh(1)x = \boxed{\cosh(1) + \sinh(1)x}$$

The remainder at *x*=0 then equals:

$$|R_1(0)| = |f[x_1, x_2, 0](0-x_1)(0-x_2)| = |(\cosh 1 - 1)(0-x_1)(0-x_2)| = |(\cosh 1 - 1)(0+1)(0-1)| = \cosh 1 - 1 = 0.543\,.$$

This of course can also be computed directly: $|R_1(0)| = |P_1(0) - f(0)| = |\cosh(1) - 1| = \cosh 1 - 1 = 0.543\,.$

It is interesting that the actual maximal error is achieved slightly to the right of the mid-point (see figure below):

$$\max_{[-1,\,1]} |R_1(x)| = 0.5576$$

Now, let's compare this with the overall error bound:

$$|R_1(x)| = \left|\frac{f''(c)}{2!}(x-x_1)(x-x_2)\right|$$

$$\le \max_{[-1,\,1]}\left|\frac{f''(c)}{2!}\right| \cdot \max_{[-1,\,1]}|(x-x_1)(x-x_2)|$$

$$= \max_{[-1,\,1]}\left|\frac{e^c}{2}\right| \cdot \max_{[-1,\,1]}|x^2-1| = \frac{e}{2}|0-1| = \frac{e}{2} = 1.359$$

As expected, the error bound is an over-estimate of the true maximal error.

Note that the first term (the derivative of exponential) is the largest at the right end of the interval, where the error is actually reduced by the nearby interpolation point!

Conclusion: exact error formula is of course more accurate, but the error bound approach gives a simple way to find the maximal error over the **entire** interval, with minimal extra computation.

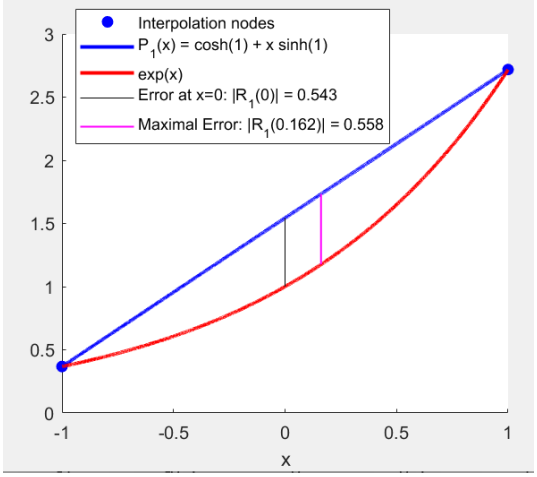


Figure 1: Linear interpolation of $exp(x)$ using nodes {-1, +1}. Note that the largest error is slightly to the right of the mid-point, $x = 0$.

Optimal Interpolation Node Choice: Chebyshev Points

From the figure above, it is obvious that endpoints (and, more generally, equidistant points) are far from an optimal choice. In fact, we proved directly in class that the basis function $|\psi_3(x)| = |(x-x_1)(x-x_2)|$ appearing in the error formula is minimized over the interval [-1, 1] when $x_{1,2} = \pm\cos\frac{\pi}{4} = \pm\frac{1}{\sqrt{2}}$. We get a much better fit with this choice, shown in the **Figure 2**. Note that the maximal error is achieved at the endpoints, not in the middle of the domain.

